

SHAWNEE, OK, USA

WMO: 722187

Lat: 35.357N

Lon: 96.943W

Elev: 1073

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 93911

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.2	20.1	4.9	7.3	20.1	9.4	9.2	26.6	27.0	46.0	24.8	45.8	7.8	10	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	100.5	75.9	98.6	76.3	95.1	76.2	80.4	92.4	79.1	91.2	77.9	90.0	9.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	148.2	86.5	75.5	139.0	83.6	74.7	135.3	83.4	45.1	92.0	43.3	91.9	42.2	89.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.3	20.9	18.9	8.8	103.0	5.1	5.1	5.2	106.7	2.2	109.6	-0.7	112.4	-4.4	116.1	
			DB												
			WB	8.0	81.5	4.9	2.4	4.5	83.2	1.6	84.6	-1.1	85.9	-4.6	87.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	62.7	40.9	44.3	54.5	61.8	70.1	79.1	82.7	82.3	75.8	63.6	53.3	42.7
	DBStd	17.16	10.79	11.42	10.40	8.86	7.48	5.22	5.32	5.97	6.93	9.25	9.04	9.70
	HDD50	967	318	224	71	13	0	0	0	0	0	10	67	264
	HDD65	3078	748	583	349	161	39	0	0	0	6	139	362	691
	CDD50	5599	36	65	210	367	624	873	1013	1001	773	433	166	38
	CDD65	2234	0	4	24	64	197	423	548	536	329	97	11	1
	CDH74	24357	11	31	197	586	1704	4521	6626	6401	3309	859	106	6
	CDH80	11603	1	4	31	147	573	2082	3546	3452	1504	252	11	0
(14) Wind	WSAvg	8.6	8.4	8.9	10.0	10.7	9.1	8.9	7.4	7.3	7.2	8.2	9.0	8.3
(15) Precipitation	PrecAvg	37.9	1.4	2.0	3.0	3.9	5.7	4.3	2.2	2.5	4.1	4.1	2.8	1.9
	PrecMax	50.4	3.7	6.3	7.3	8.3	14.4	10.8	8.5	8.5	9.3	15.0	8.2	10.7
	PrecMin	24.1	0.0	0.0	0.2	0.7	1.2	0.9	0.1	0.2	0.7	0.1	0.0	0.2
	PrecStd	7.6	1.2	1.5	1.7	1.9	3.3	2.9	1.7	1.9	2.5	3.5	2.1	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.3	78.7	83.5	90.2	92.7	99.8	107.9	108.8	100.0	90.7	81.4	72.8
		MCWB	54.1	60.7	63.6	67.9	72.4	76.1	75.7	75.1	75.9	71.9	63.9	60.9
	2%	DB	69.8	72.1	79.4	83.6	90.1	96.8	101.5	101.7	96.5	86.5	77.0	67.9
		MCWB	54.7	57.2	62.3	65.5	73.5	75.7	75.5	75.3	75.8	70.6	61.5	57.3
	5%	DB	63.3	66.6	75.1	80.8	86.3	93.0	97.4	98.7	91.3	82.0	72.4	62.8
		MCWB	52.5	55.5	60.6	64.8	71.7	76.3	76.6	76.6	74.4	67.2	60.5	53.5
	10%	DB	58.5	62.7	71.6	77.0	82.2	90.3	94.7	94.9	89.0	79.1	68.7	58.5
		MCWB	48.1	53.7	58.7	63.3	69.8	75.3	76.6	76.5	72.9	66.7	58.8	50.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.3	64.2	67.4	72.1	77.0	81.3	82.6	82.5	79.6	76.2	68.6	66.1
		MCDB	65.8	71.5	78.8	81.4	88.5	92.2	94.5	95.0	93.8	85.4	74.7	71.9
	2%	WB	58.2	61.2	65.5	70.0	75.1	79.2	80.5	80.3	77.7	73.9	66.0	60.3
		MCDB	66.0	68.1	74.6	78.7	84.6	90.9	92.0	92.7	91.7	81.2	73.9	63.5
	5%	WB	52.9	58.4	63.7	68.2	73.7	77.9	79.0	78.8	76.3	72.1	63.1	55.1
		MCDB	61.2	65.1	71.2	76.2	82.5	89.0	90.5	91.4	88.4	78.8	68.7	59.1
	10%	WB	49.0	54.0	61.3	66.4	72.3	76.7	77.7	77.5	74.8	69.4	60.1	51.6
		MCDB	57.6	62.3	68.8	74.3	80.5	87.2	89.1	90.5	85.2	75.8	66.4	56.3
(35) Mean Daily Temperature Range	5% DB	MDBR	21.0	20.6	21.7	21.8	19.2	18.9	20.2	20.6	21.2	21.8	21.9	19.3
		MCDBR	29.0	27.0	26.1	25.6	22.0	21.6	24.2	24.9	24.0	24.6	26.4	26.1
	5% WB	MCWBR	17.7	16.8	14.0	13.6	10.0	8.2	7.7	7.9	9.5	12.3	15.1	16.8
		MCWBR	26.4	24.1	20.4	21.3	18.9	18.2	19.4	21.1	21.0	19.4	22.1	20.5
(40) Clear-Sky Solar Irradiance	taub		0.297	0.302	0.343	0.378	0.413	0.425	0.429	0.435	0.388	0.346	0.317	0.294
		taud	2.504	2.509	2.418	2.334	2.303	2.309	2.334	2.264	2.373	2.483	2.505	2.522
	Ebn at Noon		289	300	294	287	277	273	271	267	275	280	279	283
		Edh at Noon	26	29	35	40	42	41	40	42	36	30	26	24
(44) All-Sky Solar Radiation	RadAvg		862	1046	1371	1698	1834	2103	2136	1941	1604	1219	909	730
	RadStd		88	159	86	123	191	212	115	119	148	159	117	86

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page